Kate J. **Davison**

07367672072 | katexxboo@gmail.com | kate-davison | kate-davison83

PhD student developing graph neural network (GNN) machine learning models to predict the 3D structures of molecules from interatomic distances. Proficient in data analysis and Python, with experience in parallel computing, GPU acceleration, and training machine learning models on high-performance computing (HPC) supercomputers.

Professional Experience

University of Bristol

Bristol, UK

Postgraduate Teaching Assistant

Sep 2025 - Current

- Supporting students in developing computational and programming skills
- Explaining technical concepts and helping troubleshoot coding and computational problems

Pharmaxo

Corsham, Bath, UK

Quality Assurance Officer

May 2024 - August 2024

- Pharmaxo produces thousands of aseptically (sterile) compounded injectable pharmaceutical products each day for hospitals and patients who are fighting cancer, living with chronic disease or need of pain relief.
- The final review of pre-manufacture batch documentation, following GMP guidelines
- The Approval for Use (final release) of all products manufactured or repackaged according to licence requirements, GMP and professional discretion
- Ensuring compliance with GMP, COSHH and Health and Safety regulations and departmental SOPS
- Supporting the QA team to include all parts of the QMS (Deviation, ICA, RCI, supplier complaints)

Bristol Royal Infirmary

Bristol, UK

Senior Assistant Technical Officer (SATO)

Sept 2023 - May 2024

- Manufacturing of aseptic products including eye drops, CIVAS, MABs and Lipid Syringes using aseptic techniques and adhering to SOPs and compliance with GMP.
- Aseptic transfer of products from clean rooms using validated aseptic technique

Computational Skills

PROGRAMMING & SOFTWARE

- Python, Bash and TypeScript.
- Scientific computing and machine learning with NumPy, SciPy, Pandas, Polars, RDKit, scikit-learn, CCDC and PyTorch.
- HPC computing on Isambard, including parallelisation and CUDA.
- Use of LLMs to support productivity and integrate AI into engineering and data workflows.

OPEN-SOURCE SOFTWARE

- EMS (Efficient Molecular Storage): package for processing chemical data and associated properties.
- inv-IMPRESSION: machine learning for predicting NMR chemical shifts and scalar couplings.

Publications

1. C. Jones, H. Cooke, **K. Davison**, M. Tilby 2026 *Organic Chemistry Frontiers* A comprehensive guide to unsupervised machine learning for understanding and developing synthetic methodology

Education

Technology Enhanced Chemical Synthesis, University of Bristol

Bristol, UK

PhD, Machine Learning, Computational Chemistry

Sep 2024 - Current

- Using Graph Neural Networks for structural elucidation of small molecules. Integrating multimodality into machine learning models.

University of Sheffield

Sheffield, UK

MChem, Chemistry

Oct 2019 - Jun 2023

- Degree classification: **II:I**

University of East Anglia (UEA)

Norwich, UK

MBBS Medicine

Oct 2018 - Jun 2019

Interests

Sport

- I am a keen runner and have ran several marathons, half marathons and ultra marathons.
- I am a keen cyclist and am preparing for a half-ironman next year.

Music

- I play regularly in a symphony orchestra and have Grade 8 Music Theory, Grade 6 Violin (Playing at grade 8 standard), Grade 5 Piano (Playing at grade 7 standard)